

Unveiling the complementary synergy of CLIP and diffusion models for weakly supervised semantic segmentation

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ABSTRACT

Weakly supervised semantic segmentation (WSSS) relies solely on image-level labels, deriving Class Activation Maps (CAMs) as pixel-level seeds and converting them into pseudo labels for segmentation. Recently, methods have leveraged either Contrastive Language-Image Pretraining (CLIP) or diffusion models to generate CAMs within WSSS pipelines. However, how to combine these two paradigms within a single framework remains underexplored. In this work, we propose **ComCD** (**Com**plementary synergy of **CLIP** and **Diffusion** models), which integrates these two paradigms and leverages their complementarity to improve CAM quality. First, class-specific CAMs are derived from the CLIP branch and the diffusion model branch. Second, we design an entropy-based fusion that maps the entropy difference between the two CAMs to reliability weights, fuses them into a refined CAM, and converts it into a pseudo mask. Third, a trainable segmentation network with a logit gating module predicts weights to fuse the two branches and produces the final segmentation. Experimental results demonstrate the superiority of the proposed ComCD over recent state-of-the-art methods in both WSSS and open-vocabulary semantic segmentation.

1. Introduction

Weakly supervised semantic segmentation (WSSS) aims to balance annotation cost with pixel-level prediction. Unlike fully supervised methods (Chen et al., 2017; Fu et al., 2025) that require pixel-level annotations, WSSS trains dense segmentation models with more economical supervision, including point annotations (Bearman et al., 2016), scribbles (Lin et al., 2016; Vernaza & Chandraker, 2017), bounding boxes (Lee et al., 2021b; Oh et al., 2021), and image-level labels (Wu et al., 2024a,b; Yang et al., 2025b), thus reducing the overhead of data collection and curation. Among these forms, image-level labels indicate only class presence without spatial localization and are therefore the most challenging. In this work, we adopt image-level labels for semantic segmentation.

Under image-level supervision, a typical multi-stage pipeline first trains an image classifier to produce class activation maps (CAMs) (Zhou et al., 2016). A refinement network is trained to further improve the CAMs (Ahn et al., 2019; Ahn & Kwak, 2018). Finally, a segmentation network is trained on pseudo labels derived from CAMs (Kweon et al., 2023; Xie et al., 2022; Yoon et al., 2024b). To reduce pipeline overhead, single-stage methods integrate these steps within one model that simultaneously generates pseudo labels and learns pixel-level masks (Ru et al.,

2023; Wu et al., 2024b; Yang et al., 2024). However, due to insufficient supervision, CAMs tend to focus only on discriminative regions, resulting in incomplete coverage and noisy pseudo labels that ultimately degrade WSSS performance. More recently, two paradigms based on Contrastive Language-Image Pretraining (CLIP) and diffusion models have gained prominence in WSSS (Lin et al., 2023; Sun et al., 2024a; Yang et al., 2025b; Yoon et al., 2024a; Zhang et al., 2024). Representative methods from each paradigm are as follows. Among CLIP-based methods, CLIP-ES (Lin et al., 2023) is a training-free, text-driven Grad-CAM that directly localizes class-discriminative regions, while ExCEL (Yang et al., 2025b) employs patch-text alignment to enhance class localization and boundary sharpness. Among diffusion-based methods, DiG (Yoon et al., 2024a) incorporates pretrained diffusion embeddings to promote region-level continuity, and iSeg (Sun et al., 2024a) iteratively refines diffusion model cross-attention to yield more spatially coherent masks.

By visualizing CLIP-based and diffusion-based CAMs (Fig. 1(a)), we observe distinct behaviors: CLIP-based methods emphasize class localization, whereas diffusion-based methods enhance spatial coherence. These observations suggest potential complementarity. To further assess this, we conduct an experiment (Fig. 1(b)) that compares the two branches' predicted segmentation masks and profiles accuracy with respect to interior distance to the ground-truth boundary. Specifically,

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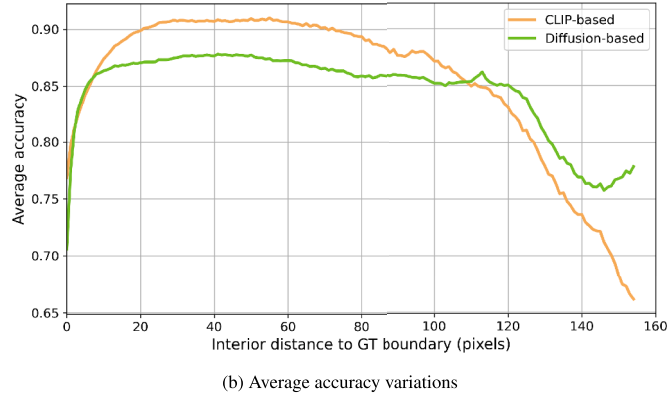
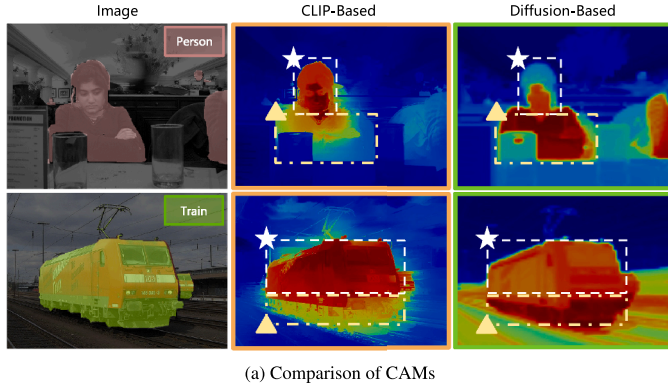


Fig. 1. Motivation for fusing CLIP-based and diffusion-based CAMs. (a) CLIP-based vs. Diffusion-based CAMs. CLIP-based CAMs highlight class localization by strongly activating class-discriminative regions. Diffusion-based CAMs favor spatial coherence, showing similar activation across pixels that belong to the same semantic region. Notation: ★ class localization, ▲ spatial coherence. (b) CLIP-based vs. Diffusion-based label accuracy. Accuracy varies with interior distance to the GT boundary (pixels).

let $\partial\Omega$ denote the ground-truth object boundary and $d(p)$ be the shortest Euclidean distance from pixel p to $\partial\Omega$; for a distance bin I and a branch $b \in \{\text{clip}, \text{diff}\}$, we define the interior-distance accuracy as $\text{Acc}_b(I) = \frac{1}{|\Omega_I|} \sum_{p \in \Omega_I} \mathbb{1}[\hat{y}_b(p) = y^*(p)]$, where $\Omega_I = \{p \mid d(p) \in I\}$, $y^*(p)$ is the ground-truth label, and $\hat{y}_b(p)$ is the prediction of branch b . The curves reveal a clear pattern: for interior pixels measured from the ground-truth boundary, CLIP-derived labels are more accurate within 100 px, while diffusion-derived labels are better beyond 100 px. Intuitively, this suggests a simple geometric heuristic: using CLIP-derived labels at boundaries and diffusion-derived labels in the interior. A naive distance-based scheme is to set a boundary-distance threshold and fuse by selecting one branch near boundaries and the other in interiors. However, object shapes are irregular and multi-scale, so using a hard threshold on pixel-level boundary distance as the fusion criterion is unreliable. Therefore, we replace distance thresholds with an uncertainty-based criterion. Uncertainty is quantified by the Shannon entropy of each pixel's class-probability distribution. Entropy is low when the predicted distribution is sharply peaked and high when it is near-uniform, so lower entropy indicates higher confidence and in turn higher pixel-level segmentation accuracy. Accordingly, at each pixel we compare the entropy values from the CLIP-derived and diffusion-derived CAMs and assign higher weight to the lower-entropy CAM, thereby avoiding the brittleness of distance-based thresholds and yielding a more reliable fusion.

In this work, we propose **ComCD** (**Complementary synergy of CLIP and Diffusion models**). ComCD adopts a simple and effective Entropy-Based Fusion (EBF) strategy within a two-branch WSSS pipeline (CLIP-based CAMs and diffusion-based CAMs). For each pixel, ComCD computes the entropy of each CAM and maps the entropy difference to a

per-pixel confidence weight that guides fusion. Applying this weight map to fuse the two CAMs yields a refined CAM, which is subsequently converted into a pseudo mask. In addition, ComCD introduces the Feature Aligned Decoder (FAD), a segmentation network trained under pseudo masks. Specifically, image embeddings from CLIP and the diffusion model are passed through a Feature Aligner that aligns them to a common spatial resolution and channel width. The aligned embeddings are then fed into a decoder, which produces segmentation logits for both branches. A Logit Gating Module (LGM) takes these two logits as input, predicts a per-pixel weight, and combines them into a fused prediction. FAD is trained under pseudo mask supervision to produce the final segmentation mask. Experiments evaluate the proposed approach under WSSS and open-vocabulary semantic segmentation settings. Under image-level supervision on PASCAL VOC 2012 and MS COCO 2014, the pseudo masks provide higher-quality supervision than either branch alone, and a decoder trained on these masks attains competitive performance. Under the open-vocabulary setting on PASCAL-Context, MS COCO-Object, and PASCAL VOC 2012, the method achieves strong results.

The main contributions are threefold:

- We propose a simple and effective Entropy-Based Fusion that converts per-pixel differences between CLIP-based and diffusion-based CAMs into reliability weights, yielding a refined CAM and pseudo masks.
- We develop the Feature Aligned Decoder with a Feature Aligner and a Logit Gating Module, trained under supervision of the pseudo mask.
- Extensive experiments demonstrate competitive performance on WSSS benchmarks (PASCAL VOC 2012, MS COCO 2014) and strong open-vocabulary results on PASCAL-Context, MS COCO-Object, and PASCAL VOC 2012.

2. Related work

Weakly Supervised Semantic Segmentation. Weakly supervised semantic segmentation (WSSS) with image-level labels commonly relies on class activation maps (CAMs) to provide dense supervision for learning a segmentation model (Lee et al., 2022a; Selvaraju et al., 2017; Wang et al., 2020; Wu et al., 2021; Zhou et al., 2016). A key limitation is that CAMs usually emphasize highly discriminative regions and thus yield incomplete object coverage, which further amplifies noise and co-occurrence bias in pseudo masks (Jiang et al., 2022; Lee et al., 2022a,b; Wang et al., 2020; Wu et al., 2024a; Yang et al., 2024). To address this issue, mainstream methods adopt multi-stage refinement, where initial cues are progressively expanded and denoised via affinity propagation, regularization, and iterative self-training (Ahn et al., 2019; Ahn & Kwak, 2018; Wei et al., 2018). To reduce the complexity of such pipelines, recent efforts increasingly shift toward single-stage WSSS, which couples CAM generation and online refinement more tightly and directly trains the segmentation head with refined pseudo labels, thereby reducing repeated retraining and improving efficiency (Ru et al., 2023; Wu et al., 2024b; Yao et al., 2026). More recently, foundation-model priors further reshape WSSS: CLIP brings strong class-aware semantics (Radford et al., 2021), while diffusion models provide structure- and shape-consistent cues (Ho et al., 2020; Rombach et al., 2022). Most existing studies exploit these priors separately, as reviewed in the following two subsections. In contrast, our work is the first to unify and jointly exploit diffusion-derived structural priors and CLIP-derived class-aware priors within a single WSSS framework.

CLIP for WSSS. CLIP (Radford et al., 2021) aligns images and texts through large-scale contrastive pretraining, which provides class-aware semantics that can guide pseudo-label generation in WSSS where only image-level labels are available. This alignment has motivated a series of CLIP-based WSSS methods (Jang et al., 2024; Lin et al., 2023; Tang et al., 2024; Xie et al., 2022; Yang et al., 2025b; Zhang et al.,

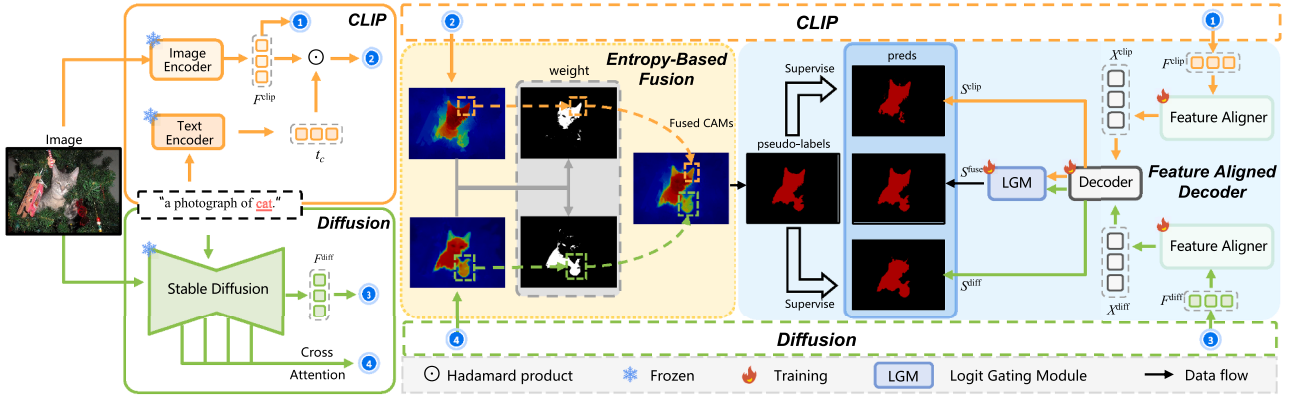


Fig. 2. ComCD overview. Given an image and class prompts, CLIP yields CLIP-based CAMs. Meanwhile, a Stable Diffusion model yields diffusion-based CAMs. An Entropy-Based Fusion computes per-pixel weights to form a fused CAM that is converted into a pseudo mask. The pseudo mask supervises the Feature Aligned Decoder (FAD). FAD uses features extracted by pretrained backbones, employs a Feature Aligner to bring the two branches to a common spatial resolution and channel width, uses a shared decoder to predict per-pixel logits for both branches, and applies a Logit Gating Module to produce the final fused prediction via learned per-pixel weights.

2024; Zhao et al., 2024) that improve localization quality and reduce background confusion. CLIMS (Xie et al., 2022) introduces language-guided calibration of CAMs with foreground and background prompts to suppress co-occurrence (Yang et al., 2024) and clean pseudo labels. CLIP-ES (Lin et al., 2023) turns frozen CLIP into a largely training-free segmenter by replacing logit-based Grad-CAM (Selvaraju et al., 2017) with probability-based Grad-CAM and refining masks using CLIP's ViT attention. QA-CLIMS (Deng et al., 2023) builds adaptive foreground and background texts via question-answer prompting to expand true object regions. WeCLIP (Zhang et al., 2024) treats frozen CLIP as a strong single-stage backbone with a lightweight decoder for efficient WSSS. ExCEL (Yang et al., 2025b) shifts from global image-text alignment to dense patch-text alignment, with text semantic enrichment and visual calibration to mine CLIP's fine-grained priors. However, CLIP-based pipelines struggle in non-discriminative regions, while diffusion models provide spatial coherence. Therefore, ComCD combines CLIP's class localization with diffusion-derived structural priors through adaptive fusion.

Diffusion model for WSSS. Diffusion models (Ho et al., 2020; Rombach et al., 2022) learn to denoise from noise to images and expose rich attention that encodes object shape and semantics. In WSSS, these attentions can be converted into CAMs or used to refine pseudo labels with only image-level labels, which has motivated several diffusion-based pipelines. DiG (Yoon et al., 2024a) proposes Locality Fusion Cross-Attention that fuses pretrained diffusion model embeddings with a ViT-based WSSS model using only image-level labels. SeeDiff (Park et al., 2025) uses cross-attention as seeds and expands them with multi-scale self-attention to generate high quality masks off-the-shelf, without extra training, prompt tuning, or a pretrained segmentation network. Diff-Segmenter (Wang et al., 2025) reveals that self-attention captures object shape while cross-attention indicates semantics. iSeg (Sun et al., 2024a) introduces iterative refinement with an entropy-reduced self-attention module that progressively improves cross-attention maps for training-free segmentation. Conversely, diffusion-only pipelines favor spatial coherence but offer limited class specificity. ComCD complements diffusion-only pipelines with CLIP's class localization and fuses the two branches using entropy-based weighting.

3. Methodology

ComCD leverages pretrained CLIP and a text-conditioned diffusion model to obtain CAMs, fuses them into pseudo masks, and then trains a decoder under these pseudo masks to predict the final segmentation masks. Section 3.1 reviews how CAMs are obtained using CLIP and the diffusion model. Section 3.2 introduces an Entropy-Based Fusion that

maps entropy differences to weights, fusing the two CAMs into a pseudo mask (Fig. 2, Entropy-Based Fusion). Section 3.3 presents the Feature-Aligned Decoder, a shared decoder trained on the pseudo mask, which employs a Logit Gating Module to weight the two branch logits and produce the final segmentation (Fig. 2 Feature Aligned Decoder).

3.1. Preliminaries

3.1.1. Diffusion-based class activation maps

ComCD uses a pretrained and frozen text-conditioned diffusion model (Stable Diffusion Rombach et al., 2022). Given an image and its class prompt, ComCD encodes the image into the latent space, adds Gaussian noise at a fixed timestep, and performs a single reverse denoising step with a fixed sampling configuration to extract self-attention and cross-attention maps. These diffusion hyperparameters follow the publicly released iSeg (Sun et al., 2024a) configuration and are kept fixed across all experiments. Because the diffusion backbone is frozen and both the timestep and sampling configuration are fixed during CAM generation, the extracted attention maps and the resulting diffusion-based CAMs are deterministic for a given input image. Cross-attention between latent image embeddings and text embeddings in this step yields initial CAMs. In addition, self-attention from the same denoising step provides spatial affinity and is used to iteratively propagate and refine these maps, obtaining refined diffusion-based CAMs.

Cross-attention map. Given the image and the corresponding class prompt, a single reverse denoising step at timestep t in a pretrained and frozen text-conditioned diffusion model yields an image embedding, which is flattened to $E_{\text{img}} \in \mathbb{R}^{HW \times C}$, where H and W denote spatial height and width and C is the channel dimension. The prompt is encoded into text embeddings $E_{\text{txt}} \in \mathbb{R}^{L \times d}$, where L is the number of text tokens and d is the token embedding dimension. Learned projections $W_q^{\text{ca}}, W_k^{\text{ca}}, W_v^{\text{ca}}$ produce queries, keys, and values. Queries come from image embeddings $Q^{\text{ca}} = E_{\text{img}} W_q^{\text{ca}} \in \mathbb{R}^{HW \times d}$. Keys and values come from text $K^{\text{ca}} = E_{\text{txt}} W_k^{\text{ca}} \in \mathbb{R}^{L \times d}$ and $V^{\text{ca}} = E_{\text{txt}} W_v^{\text{ca}} \in \mathbb{R}^{L \times d}$. The cross-attention map is

$$A^{\text{ca}} = \text{Softmax}\left(\frac{Q^{\text{ca}} K^{\text{caT}}}{\sqrt{d}}\right) \in \mathbb{R}^{HW \times L}. \quad (1)$$

Following iSeg (Sun et al., 2024a), we collect cross-attention maps from several spatial scales, apply softmax along the token dimension, combine them with fixed weights, and bilinearly upsample them to a common low-resolution grid; all later refinement steps operate on this aggregated cross-attention map. For each semantic class, we precompute the token indices corresponding to its class name and, for class c , average

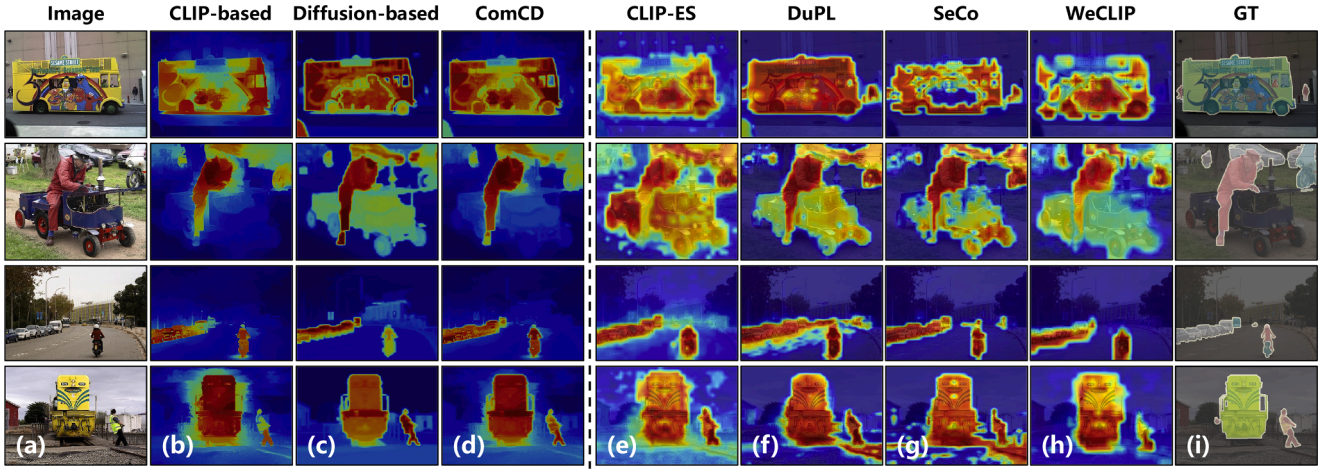


Fig. 3. Qualitative comparison of CAMs. For each image, CAMs activate all present classes. (a) Image. (b) CLIP-based CAMs. (c) Diffusion-based CAMs. (d) ComCD. (e) CLIP-ES (Lin et al., 2023). (f) DuPL (Wu et al., 2024b). (g) SeCo (Yang et al., 2024). (h) WeCLIP (Zhang et al., 2024). (i) Ground truth.

the selected columns of the aggregated cross-attention and reshape the result into a class-aware spatial map $a_c^{ca} \in \mathbb{R}^{H_{low} \times W_{low}}$.

Self-attention map. For self-attention over the image embeddings, learned projections $W_q^{sa}, W_k^{sa}, W_v^{sa}$ map E_{img} to queries, keys, and values: $Q^{sa} = E_{img} W_q^{sa}, K^{sa} = E_{img} W_k^{sa}, V^{sa} = E_{img} W_v^{sa}, Q^{sa}, K^{sa}, V^{sa} \in \mathbb{R}^{HW \times d}$. The self-attention map is

$$A^{sa} = \text{Softmax}\left(\frac{Q^{sa} K^{saT}}{\sqrt{d}}\right) \in \mathbb{R}^{HW \times HW}. \quad (2)$$

In practice, we take the $H_{low} \times W_{low}$ latent resolution so that the spatial indices of A^{sa} are aligned with those of a_c^{ca} ; this attention captures long-range spatial affinity and favors region connectivity.

Iterative refinement with self-attention. For class token c , the aggregated cross-attention map described above provides an initial class-specific map a_c^{ca} that highlights spatial regions most relevant to class c on the 64×64 grid. Following iSeg (Sun et al., 2024a), we then use the self-attention as a spatial affinity to refine this map in a single propagation step. Concretely, a_c^{ca} is vectorized as $m_c \in \mathbb{R}^{H_{low} W_{low}}$, and the self-attention at the same resolution is reshaped into a spatial-spatial affinity matrix $A^{sa} \in \mathbb{R}^{H_{low} W_{low} \times H_{low} W_{low}}$. We then compute

$$\tilde{m}_c = A^{sa} m_c, \quad (3)$$

which diffuses class scores along high-affinity spatial neighbors. Finally, $\tilde{m}_c$ is reshaped back to $\mathbb{R}^{H_{low} \times W_{low}}$, bilinearly upsampled to (H, W) , and normalized to obtain the refined diffusion-based CAM $\mathcal{H}_c^{diff} \in \mathbb{R}^{H \times W}$. This refinement step improves spatial coherence and suppresses spurious isolated responses without introducing additional trainable parameters.

3.1.2. CLIP-based class activation maps

CLIP (Radford et al., 2021) aligns images and text by contrastive pretraining on large-scale image-text pairs. It consists of an image encoder (typically a Vision Transformer Dosovitskiy et al., 2020) and a text encoder, both of which project inputs into a shared embedding space where cosine similarity reflects semantic alignment.

An input image is fed to the CLIP image encoder to obtain patch embeddings $E_{img} \in \mathbb{R}^{HW \times D}$ from selected layers. Linear projections W_Q, W_K, W_V yield $Q = E_{img} W_Q, K = E_{img} W_K$, and $V = E_{img} W_V$. For notational simplicity, we omit the attention-head dimension and write $Q, K, V \in \mathbb{R}^{HW \times D}$, where D is the projected channel width.

Unlike standard self-attention, where Q and K arise from different projections W_Q and W_K , Q is reused for both query and key, i.e., $Q^{sa} = K^{sa} = Q$, and the attention takes the form $A_{QQ} = \text{Softmax}(QQ^T / \sqrt{D}) \in \mathbb{R}^{HW \times HW}$. Analogously, replacing (Q^{sa}, K^{sa}) with (K, K) and (V, V)

yields A_{KK} and A_{VV} . ComCD defines the spatial affinity as $\mathcal{A} = \frac{1}{3}(A_{QQ} + A_{KK} + A_{VV})$. Applying $\mathcal{A}$ to the values V produces $\tilde{E}^{img} = \mathcal{A}V$. ComCD then reshapes $\tilde{E}^{img}$ to $\hat{E}^{img} \in \mathbb{R}^{H \times W \times D}$ to obtain per-pixel embeddings. Finally, given the text embedding $E_c^{txt} \in \mathbb{R}^D$ for class c , the CAM is computed by cosine similarity:

$$\mathcal{H}_c^{clip} = \cos\left(\frac{\hat{E}^{img}}{\|\hat{E}^{img}\|_2}, \frac{E_c^{txt}}{\|E_c^{txt}\|_2}\right) \in \mathbb{R}^{H \times W}, \quad (4)$$

where $\cos(\cdot, \cdot)$ denotes cosine similarity.

3.1.3. Diffusion-based CAM vs. CLIP-based CAM

The previous two paragraphs describe how diffusion-based CAMs and CLIP-based CAMs are obtained in our framework. By visualizing them in Fig. 1(a), we observe distinct behaviors: diffusion-based CAMs favor *spatial coherence*, yielding smoother and more region-complete activations that better preserve object structure (Sun et al., 2024a), whereas CLIP-based CAMs favor *class localization*, strongly highlighting class-discriminative parts and producing sharper responses around object boundaries, while remaining sparse on non-discriminative regions (Yang et al., 2025b).

Regarding the commonly used CAM acquisition methods, CLIP-based CAMs are typically obtained by attributing an image-text similarity score to spatial tokens, whereas diffusion-based CAMs are usually extracted from the text-conditioned diffusion U-Net attentions.

3.2. Entropy-based fusion

Building on Section 3.1, ComCD constructs an Entropy-Based Fusion that combines the CLIP-based CAMs $\{\mathcal{H}_c^{clip}\}_{c \in C_{img}}$ and the diffusion-based CAMs $\{\mathcal{H}_c^{diff}\}_{c \in C_{img}}$. As observed in Fig. 1(b), CLIP-based CAMs tend to be more precise around object boundaries, whereas diffusion-based CAMs are more spatially coherent in object interiors; EBF is therefore designed as a per-pixel rule that decides which branch is more reliable at each location and weights their contributions accordingly. Given an input image, let C_{img} denote the set of classes present in the image ($|C_{img}|$ is the number of classes). Stacking the per-class CAMs yields $\mathcal{H}^b \in \mathbb{R}^{|C_{img}| \times H \times W}$ for each branch $b \in \{clip, diff\}$. As the first step, each branch is converted into a pixel-level class distribution by applying a softmax along the class dimension:

$$\mathcal{P}_c^b = \frac{\exp(\mathcal{H}_c^b)}{\sum_{c' \in C_{img}} \exp(\mathcal{H}_{c'}^b)}. \quad (5)$$

At each spatial location (h, w) , the entropy is defined as the Shannon entropy of the per-pixel class probability distribution induced by branch

$b \in \{\text{clip}, \text{diff}\}$, namely $\{\mathcal{P}_{c,h,w}^b\}_{c \in C_{\text{img}}}$:

$$e_{h,w}^b = - \sum_{c \in C_{\text{img}}} \mathcal{P}_{c,h,w}^b \log \mathcal{P}_{c,h,w}^b. \quad (6)$$

where $\sum_{c \in C_{\text{img}}} \mathcal{P}_{c,h,w}^b = 1$ holds for every (h, w) . Predictive entropy is a standard uncertainty measure for softmax-based classifiers: for a fixed label set, lower entropy corresponds to a more peaked distribution, where one class dominates the others. In practice, such low-entropy predictions are empirically associated with higher accuracy in both classification and segmentation tasks. In ComCD, we exploit this property in a relative sense: when the CLIP and diffusion branches are compared on the same pixel, the branch with lower entropy is treated as more reliable. This design is consistent with the boundary–interior curves in Fig. 1(b), where the CLIP branch is both more confident and more accurate near object boundaries, whereas the diffusion branch is more confident and more accurate in object interiors. Compared with alternative confidence proxies such as maximum probability, margin, or variance, entropy has the practical advantage that it is computed directly from a single per-pixel softmax distribution and provides a monotonic scalar summary of distribution sharpness regardless of the number of classes. In our pseudo-mask generation stage with frozen CLIP and diffusion backbones, this makes entropy a particularly convenient and lightweight choice.

Based on this relative-confidence view, the difference $\Delta e_{h,w} = e_{h,w}^{\text{clip}} - e_{h,w}^{\text{diff}}$ is computed and used as a per-pixel reliability criterion. Lower entropy indicates higher confidence. Therefore, $\Delta e_{h,w} < 0$ assigns greater confidence to CLIP at (h, w) , whereas $\Delta e_{h,w} > 0$ favors the diffusion model branch. The quantity Δe is passed through a sigmoid to obtain a per-pixel weight map $W = \sigma(\Delta e)$, with $\sigma(\cdot)$ denoting the sigmoid function. The fused mask is defined via element-wise multiplication as

$$\mathcal{H}_c^{\text{fuse}} = W \odot \mathcal{P}_c^{\text{diff}} + (\mathbf{1} - W) \odot \mathcal{P}_c^{\text{clip}}. \quad (7)$$

where $\odot$ denotes element-wise multiplication and $\mathbf{1} \in \mathbb{R}^{H \times W}$ is the all-ones map, i.e., every entry equals 1. This expression yields the fused map for class c over all pixels.

The final mask $\mathcal{M}$ over the image-specific class set C_{img} is then obtained by a pixel-level argmax:

$$\mathcal{M} = \arg \max_{c \in C_{\text{img}}} \mathcal{H}_c^{\text{fuse}}. \quad (8)$$

In addition to being directly evaluated as a pseudo mask, the final mask $\mathcal{M}$ can also serve as supervision to train the segmentation network.

3.3. Feature aligned decoder

To further explore the potential of CLIP-diffusion fusion, a segmentation network is trained under the supervision of the pseudo mask $\mathcal{M}$ obtained from the entropy-based CAM fusion in Section 3.2. For each branch, embeddings are gathered from different blocks of its respective model and aligned by a Feature Aligner (FA) module so that a single decoder can be shared.

Let $\{F_i^b\}_{i=1}^{T_b}$ denote embeddings gathered from different blocks of branch $b \in \{\text{clip}, \text{diff}\}$. Here T_b is the number of selected blocks from branch b . FA upsamples each F_i^b to (H_d, W_d) , projects channels to a shared width D , concatenates them along channels, and applies a 1×1 convolution to form the decoder input:

$$X^b = \text{Conv}_{1 \times 1} \left(\text{Cat}_{i=1}^{T_b} (\text{Up}(F_i^b)) \right) \in \mathbb{R}^{B \times D \times H_d \times W_d}, \quad (9)$$

where $\text{Up}(\cdot)$ upsamples each embedding to (H_d, W_d) , $\text{Cat}(\cdot)$ denotes channel-wise concatenation, and $\text{Conv}_{1 \times 1}(\cdot)$ is a 1×1 convolution that projects the concatenated embedding to D channels. A shared decoder then takes X^b as input and produces per-pixel class logits $S^b \in \mathbb{R}^{B \times |C| \times H_d \times W_d}$ for each branch $b \in \{\text{clip}, \text{diff}\}$. Here C denotes the set of all categories in the dataset ($|C|$ is the number of classes).

Subsequently, a Logit Gating Module predicts a per-pixel weight W' from the predicted logits S^b :

$$W' = \sigma \left(\text{Conv}_{1 \times 1} \left(\text{Cat}(S^{\text{clip}}, S^{\text{diff}}) \right) \right) \in \mathbb{R}^{B \times 1 \times H_d \times W_d}, \quad (10)$$

where $\text{Conv}_{1 \times 1}(\cdot)$ is a 1×1 convolution mapping $2|C|$ channels to one channel, and σ is the sigmoid. Finally, consistent with the pixel-level fusion rule in Eq. (7), define the fused logits as

$$S^{\text{fuse}} = W' \odot S^{\text{diff}} + (\mathbf{1} - W') \odot S^{\text{clip}}, \quad (11)$$

where $\odot$ denotes element-wise multiplication and $\mathbf{1} \in \mathbb{R}^{H \times W}$ is the all-ones map.

3.4. Training objectives

The fused logits are supervised by the pseudo mask $\mathcal{M}$ using per-pixel cross-entropy:

$$\mathcal{L}_{\text{fuse}} = \text{CE}(S^{\text{fuse}}, \mathcal{M}), \quad (12)$$

where $\text{CE}(\cdot, \cdot)$ denotes the standard per-pixel cross-entropy. Similarly, ComCD supervises each branch with the same cross-entropy loss:

$$\mathcal{L}_{\text{clip}} = \text{CE}(S^{\text{clip}}, \mathcal{M}), \quad \mathcal{L}_{\text{diff}} = \text{CE}(S^{\text{diff}}, \mathcal{M}). \quad (13)$$

In addition, following ExCEL (Yang et al., 2025b), ComCD adopts a diversity loss, denoted by $\mathcal{L}_{\text{div}}$, to discourage redundant predictions and encourage complementary cues between the two branches. The overall training objective is

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{fuse}} + \lambda_2 \mathcal{L}_{\text{clip}} + \lambda_3 \mathcal{L}_{\text{diff}} + \lambda_4 \mathcal{L}_{\text{div}}, \quad (14)$$

where $\lambda_1, \lambda_2, \lambda_3, \lambda_4 \geq 0$ are the weights rescaling the contribution of each loss term.

4. Experiments

4.1. Experimental settings

Datasets and Evaluation Metrics. We evaluate our proposed method on four datasets: PASCAL VOC 2012 (Everingham et al., 2010), MS COCO 2014 (Lin et al., 2014), PASCAL-Context (Mottaghi et al., 2014), and MS COCO-Object (Lin et al., 2014). For PASCAL VOC 2012, which contains 21 semantic categories, ComCD follows the common practice (Long et al., 2015; Ru et al., 2023; Wu et al., 2024b) of using the extended version with the SBD dataset, consisting of 10,582 training images, 1,449 validation images, and 1,456 test images. For MS COCO 2014, which includes 81 classes, the dataset is divided into 82,081 training images and 40,137 validation images. For PASCAL-Context, which includes 60 classes, the dataset comprises 4,998 training images and 5,105 validation images. For MS COCO-Object, which includes 81 classes, ComCD evaluates on 4,952 validation images from the COCO 2014 val set. The primary evaluation metric is mean Intersection-over-Union (mIoU).

Implementation Details. ComCD uses CLIP with a ViT-B/16 (Dosovitskiy et al., 2020) image encoder and a pretrained Stable Diffusion v2.1 (Rombach et al., 2022). Both backbones remain frozen throughout training. On the CLIP side, we follow ExCEL (Yang et al., 2025b): patch-token features from all 12 transformer layers of ViT-B/16 are projected to 256 dimensions and fused with a SegFormer-style head into a unified 256-channel feature map, and class names are rendered with the fixed prompt template “a clean origami { }.” before being passed to the text encoder. On the diffusion side, we follow iSeg (Sun et al., 2024a): the diffusion scheduler is set to 1000 denoising steps, we add Gaussian noise at a timestep t uniformly sampled during training and fix $t=100$ at inference, and perform a single reverse denoising step to extract cross- and self-attention maps, which are aggregated onto a shared 64×64 latent grid (i.e., $H_{\text{low}}=W_{\text{low}}=64$) for CAM generation, with self-attention refinement iterations configured as in Sun et al. (2024a). The two branches are aligned to a common 256-channel, shared-resolution feature space and then fed into a lightweight Transformer-based decoder



Fig. 4. Visualization of WSSS pseudo masks. We compare the results of ComCD with those of ExCEL (Yang et al., 2025b), iSeg (Sun et al., 2024a), and CLIP-ES (Lin et al., 2023) on the PASCAL VOC 2012 and MS COCO 2014 datasets.

Table 1

Pseudo mask evaluation. Segmentation mIoU on PASCAL VOC 2012 train and MS COCO 2014 train.

Method	Publication	Type	Net.	VOC	COCO
Training-required methods.					
IRN (Ahn et al., 2019)	CVPR'2019	Multi-stage	ResNet50	66.5	42.4
AdvCAM (Lee et al., 2021a)	CVPR'2021	Multi-stage	ResNet101	55.6	35.8
CLIMS (Xie et al., 2022)	CVPR'2022	Multi-stage	ResNet50	56.6	–
HSC (Wu et al., 2023)	IJCAI'2023	Multi-stage	ResNet101	71.8	–
ToCo (Ru et al., 2023)	CVPR'2023	Single-stage	ViT-B/16	72.2	–
BAS (Zhai et al., 2024)	IJCV'2024	Multi-stage	ResNet50	57.7	36.9
WeCLIP (Zhang et al., 2024)	CVPR'2024	Single-stage	ViT-B/16	75.4	–
MCTformer+ (Xu et al., 2024)	TIPAMI'2024	Multi-stage	ResNet38	68.8	–
DiG (Yoon et al., 2024a)	ECCV'2024	Multi-stage	DeiT-S	74.3	–
S2C (Kweon & Yoon, 2024)	CVPR'2024	Multi-stage	WRResNet38	81.7	–
PRCE (Xu et al., 2025)	CVPR'2025	Single-stage	ResNet50	77.6	–
Training-free methods.					
DiffSegmenter (Wang et al., 2025)	TIP'2025	Single-stage	U-Net	70.5	–
T2M (Xiao et al., 2024)	Neurocomputing'2024	Single-stage	U-Net	72.7	43.7
CLIP-ES (Lin et al., 2023)	CVPR'2023	Multi-stage	ViT-B/16	70.8	39.7
iSeg (Sun et al., 2024a)	arXiv'2024	Single-stage	U-Net	75.2	45.5
ExCEL (Yang et al., 2025b)	CVPR'2025	Single-stage	ViT-B/16	74.6	43.6
Ours	–	Single-stage	ViT-B/16	82.1	51.3

head (Zhang et al., 2024). Following prior works (Wu et al., 2024b; Yang et al., 2024, 2025b), AdamW optimizes all trainable components with a learning rate of 1×10^{-4} and a weight decay of 1×10^{-2} . The loss weights are set to $\lambda_1 = 1$ and $\lambda_2 = \lambda_3 = \lambda_4 = 0.1$. Training runs for 30,000 iterations on PASCAL VOC 2012 and 100,000 iterations on MS COCO 2014. During inference, ComCD adopts the standard approach used in WSSS by applying multi-scale testing and post-processing with dense CRF.

Evaluation protocol. For WSSS on PASCAL VOC 2012 and MS COCO 2014, we first evaluate ComCD with frozen CLIP and diffusion backbones: we generate CLIP-based and diffusion-based CAMs on the train set, fuse them via the entropy-based fusion to obtain pseudo masks, and compute mIoU against the ground-truth masks on train. No additional parameters, including the decoder, are trained in this step, and the results are reported in Table 1. In the second step, these pseudo masks are used to supervise the segmentation decoder of our single-stage WSSS framework, which is trained on train and evaluated on val; the corresponding mIoU is summarized in Table 2 and compared with recent training-required WSSS methods under identical splits and metrics.

For open-vocabulary semantic segmentation, ComCD similarly adopts a pseudo mask evaluation with frozen backbones on the val

sets of PASCAL VOC 2012, PASCAL-Context, and MS COCO-Object: the frozen CLIP and diffusion backbones plus the entropy-based fusion directly produce masks, and we compute mIoU on the val sets as reported in Table 3. A pretrained CLIP model is used, and image-level class tags produced by TagCLIP (Lin et al., 2024) serve as text prompts. Unless otherwise stated, all other settings follow those used for WSSS.

4.2. Experimental results

CAMs of Fusion. Fig. 3 illustrates ComCD on PASCAL VOC 2012 alongside representative baselines. The CLIP-based CAMs in Fig. 3(b) provide class priors for localization, delineating boundaries and highlighting discriminative regions, whereas the diffusion-based CAMs in Fig. 3(c) contribute structural priors that promote regional continuity and yield more complete activations. In addition, the entropy-based per-pixel weight down-weights the less reliable branch, enabling mutual correction of erroneous activations. Our entropy-guided fusion in Fig. 3(d) integrates these strengths into a single map that aligns more closely with the ground truth in Fig. 3(i) by suppressing background leakage near edges and closing interior gaps. Compared with CLIP-ES (Lin et al., 2023),

Table 2

Weakly supervised semantic segmentation results. Segmentation mIoU on PASCAL VOC 2012 val and MS COCO 2014 val. **Sup.** denotes supervision types (*I* image-level, *S* saliency, *T* text). ‘-’ indicates not reported.

Method	Publication	Sup.	Net.	VOC	COCO
Multi-stage WSSS methods.					
L2G (Jiang et al., 2022)	CVPR’2022	<i>I</i> + <i>S</i>	ResNet101	72.1	44.2
RCA (Zhou et al., 2022b)	CVPR’2022	<i>I</i> + <i>S</i>	ResNet38	72.2	36.8
OCR (Cheng et al., 2023)	CVPR’2023	<i>I</i>	ResNet38	72.7	42.5
BECO (Rong et al., 2023)	CVPR’2023	<i>I</i>	ResNet101	73.7	45.1
MCTformer+ (Xu et al., 2024)	TIPAMI’2024	<i>I</i>	ResNet38	74.0	45.2
CTI (Yoon et al., 2024b)	CVPR’2024	<i>I</i>	ResNet101	74.1	45.4
MuP-VSS (Duan et al., 2025)	CVPR’2025	<i>I</i>	ResNet38	73.6	46.6
CLIMS (Xie et al., 2022)	CVPR’2022	<i>I</i> + <i>T</i>	ResNet50	70.4	-
CLIP-ES (Lin et al., 2023)	CVPR’2023	<i>I</i> + <i>T</i>	ResNet101	72.2	45.4
PSDPM (Zhao et al., 2024)	CVPR’2024	<i>I</i> + <i>T</i>	ResNet101	74.1	47.2
CPAL (Tang et al., 2024)	CVPR’2024	<i>I</i> + <i>T</i>	ResNet101	74.5	46.8
DiG (Yoon et al., 2024a)	ECCV’2024	<i>I</i>	WRResNet34	73.9	45.5
S2C (Kweon & Yoon, 2024)	CVPR’2024	<i>I</i>	WRResNet34	78.2	49.8
Single-stage WSSS methods.					
AFA (Ru et al., 2022)	CVPR’2022	<i>I</i>	MiT-B/16	66.0	38.9
TSCD (Xu et al., 2023b)	AAAI’2023	<i>I</i>	MiT-B/16	67.3	40.1
ToCo (Ru et al., 2023)	CVPR’2023	<i>I</i>	ViT-B/16	71.1	42.3
DuPL (Wu et al., 2024b)	CVPR’2024	<i>I</i>	ViT-B/16	73.3	44.6
SeCo (Yang et al., 2024)	CVPR’2024	<i>I</i>	ViT-B/16	74.0	46.7
PRCE (Xu et al., 2025)	CVPR’2025	<i>I</i>	ViT-B/16	75.5	47.2
DIAL (Jang et al., 2024)	ECCV’2024	<i>I</i> + <i>T</i>	ViT-B/16	74.5	44.4
WeCLIP (Zhang et al., 2024)	CVPR’2024	<i>I</i> + <i>T</i>	ViT-B/16	76.4	47.1
ExCEL (Yang et al., 2025b)	CVPR’2025	<i>I</i> + <i>T</i>	ViT-B/16	77.2	49.3
Ours	-	<i>I</i> + <i>T</i>	ViT-B/16	79.5	52.1

Table 3

Comparison with open vocabulary segmentation approaches. We report mIoU on PASCAL VOC 2012 val, PASCAL-VOC Context val, and MS COCO-Object val. ‘-’ indicates not reported.

Method	Publication	VOC	Context	Object
Training-required methods.				
ReCo (Shin et al., 2022)	NeurIPS’22	25.1	19.9	15.7
MaskCLIP (Zhou et al., 2022a)	ECCV’22	38.8	23.6	20.6
SegCLIP (Luo et al., 2023)	ICML’23	52.6	24.7	26.5
CLIPpy (Ranasinghe et al., 2023)	ICCV’23	52.2	-	32.0
ViewCo (Ren et al., 2026)	ICLR’23	52.4	23.0	23.5
OVSegmenter (Xu et al., 2023a)	CVPR’23	53.8	20.4	25.1
TCL (Cha et al., 2023)	CVPR’23	51.2	24.3	30.4
SAM-CLIP (Wang et al., 2024)	CVPRW’24	60.6	29.2	31.5
LPOSS (Stojnić et al., 2025)	CVPR’25	62.4	34.3	35.4
Training-free methods.				
TagCLIP (Lin et al., 2024)	AAAI’24	64.8	-	-
CaR (Sun et al., 2024b)	CVPR’24	67.6	30.5	36.6
DiffSegmenter (Wang et al., 2025)	TIP’25	60.1	27.5	37.9
iSeg (Sun et al., 2024a)	arXiv’24	68.2	30.9	38.4
ProxyCLIP (Lan et al., 2024)	ECCV’24	61.3	35.3	37.5
ResCLIP (Yang et al., 2025a)	CVPR’25	61.1	33.5	35.0
CASS (Kim et al., 2025)	CVPR’25	65.8	36.7	37.8
Ours	-	74.2	54.8	39.3

DuPL (Wu et al., 2024b), SeCo (Yang et al., 2024), and WeCLIP (Zhang et al., 2024) in Fig. 3(e)–(h), ComCD produces cleaner responses along object boundaries and in non-discriminative regions.

WSSS Pseudo Mask Evaluation. Table 1 reports mIoU of the generated pseudo masks on PASCAL VOC 2012 and MS COCO 2014. With ViT-B/16, ComCD attains **82.1%** on VOC and **51.3%** on MS COCO 2014. Compared with recent strong baselines in Table 1, ComCD exceeds PRCE (77.6%) by +4.5% on VOC and surpasses IRN (42.4%) on MS COCO 2014 by +8.9%. Relative to iSeg, the gains are +6.9% on VOC and +5.8% on MS COCO 2014; relative to T2M, +9.4% and +7.6%; relative to CLIP-ES, +11.3% and +11.6%; relative to ExCEL, +7.5% and +7.7%; and relative to DiffSegmenter, +11.6% on VOC. Fig. 4 provides qualitative comparisons with ExCEL (Yang et al., 2025b), iSeg (Sun et al., 2024a), and CLIP-ES (Lin et al., 2023), where ComCD yields pseudo masks with more complete object coverage and sharper boundaries while reducing false

positives, background leakage, and fragmented regions on both datasets. All differences are computed directly from Table 1. Methods without MS COCO 2014 results are compared on VOC only. The per-class mIoU radar plot in Fig. 6 further shows that ComCD surpasses CLIP-ES, ExCEL, and iSeg across all categories.

WSSS Final Segmentation Results. Table 2 summarizes segmentation mIoU on PASCAL VOC 2012 val set and MS COCO 2014 val set for multi-stage and single-stage training-required WSSS. Using *I* + *T* supervision with a ViT-B/16 backbone, our method attains **79.5%** on VOC and **52.1%** on MS COCO 2014. Within single-stage approaches, it exceeds ExCEL (Yang et al., 2025b) (77.2% and 49.3%) by +2.3% and +2.8%. Relative to leading multi-stage methods, it is higher than CPAL (Tang et al., 2024) on VOC (74.5%) by +5.0% and higher than PSDPM (Zhao et al., 2024) on MS COCO 2014 (47.2%) by +4.9%. Across other representative single-stage methods, the absolute improvements range from +1.1% to +13.5% on VOC and from +1.8% to +13.2% on MS COCO 2014. Fig. 5 provides qualitative comparisons with ExCEL (Yang et al., 2025b), DuPL (Wu et al., 2024b), and ToCo (Ru et al., 2023), and the visual results are consistent with the quantitative gains reported in Table 2.

OVSS Pseudo Mask Evaluation. Table 3 reports mIoU of the generated pseudo masks on PASCAL VOC 2012, PASCAL-Context, and MS COCO-Object. With ViT-B/16, ComCD achieves **74.2%** on VOC, **54.8%** on Context, and **39.3%** on MS COCO-Object, ranking first across all three benchmarks. Compared with LPOSS (Stojnić et al., 2025) (62.4%, 34.3%, 35.4%), the gains are +11.8%, +20.5%, and +3.9%. Relative to iSeg (Sun et al., 2024a) (68.2%, 30.9%, 38.4%), the improvements are +6.0%, +23.9%, and +0.9%, and over CASS (Kim et al., 2025) (65.8%, 36.7%, 37.8%) they are +8.4%, +18.1%, and +1.5%. The improvement on PASCAL-Context is particularly notable, while VOC shows a clear advantage and COCO-Object exhibits a smaller yet consistent margin. Qualitative comparisons of the resulting pseudo masks are shown in Fig. 7, where ComCD is contrasted with DiffSegmenter (Wang et al., 2025) and iSeg (Sun et al., 2024a). All differences are computed directly from Table 3.

Comparison with Fully-Supervised Methods. Table 4 compares weakly supervised methods with their fully supervised counterparts on the PASCAL VOC 2012 val set. Our training-required method with

Table 4

Comparisons with fully/weakly supervised methods on VOC val. F : fully-supervised. ViT-B*: pretrained from CLIP. Ratio is relative to the fully-supervised counterpart with the same backbone.

Methods	Publication	Sup.	Net.	Val	Ratio
Fully-supervised methods					
DeepLabV2 (Chen et al., 2017)	TPAMI'2017	F	ResNet101	77.7	–
DeepLabV2 (Chen et al., 2017)	TPAMI'2017	F	ViT-B/16	82.3	–
WeCLIP-Full (Zhang et al., 2024)	CVPR'2024	F	ViT-B*	81.6	–
Multi-stage WSSS methods					
CLIMS (Xie et al., 2022)	CVPR'2022	$I + T$	ResNet101	70.4	90.6%
CLIP-ES (Lin et al., 2023)	CVPR'2023	$I + T$	ResNet101	72.2	92.9%
CPAL (Tang et al., 2024)	CVPR'2024	$I + T$	ResNet101	74.5	95.9%
Single-stage WSSS methods					
ToCo (Ru et al., 2023)	CVPR'2023	I	ViT-B/16	71.1	86.4%
DuPL (Wu et al., 2024b)	CVPR'2024	I	ViT-B/16	73.3	89.1%
SeCo (Yang et al., 2024)	CVPR'2024	I	ViT-B/16	74.0	89.9%
DIAL (Jang et al., 2024)	ECCV'2024	$I + T$	ViT-B/16	74.5	90.5%
WeCLIP (Zhang et al., 2024)	CVPR'2024	$I + T$	ViT-B*	76.4	93.6%
ExCEL (Yang et al., 2025b)	CVPR'2025	$I + T$	ViT-B*	77.2	94.6%
Ours	–	$I + T$	ViT-B*	79.5	97.4%

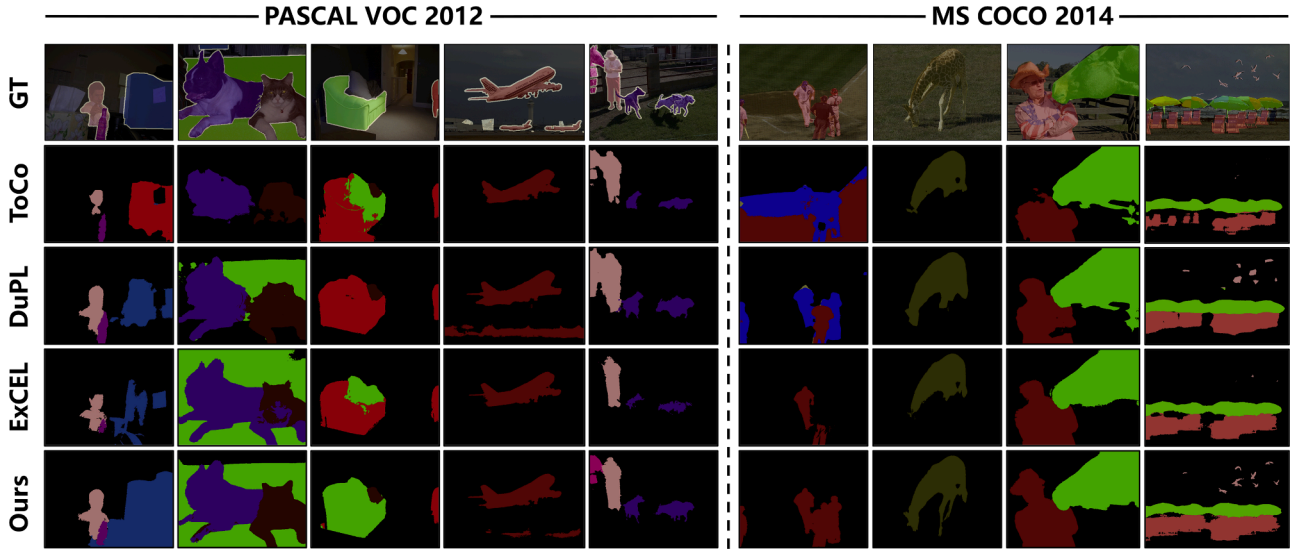


Fig. 5. Visualization of training-required weakly supervised segmentation results. We compare the results of ComCD with those of ExCEL (Yang et al., 2025b), DuPL (Wu et al., 2024b) and ToCo (Ru et al., 2023).

Table 5

Ablation Study. “Seg.” represents segmentation performance.

$\mathcal{L}_{\text{fuse}}$	$\mathcal{L}_{\text{clip}}$	$\mathcal{L}_{\text{diff}}$	$\mathcal{L}_{\text{div}}$	Seg.
✓				74.1
✓	✓			77.4
✓		✓		76.5
✓			✓	76.0
✓	✓		✓	78.5
✓		✓	✓	77.4
✓	✓	✓	✓	79.5

ViT-B* achieves 79.5% mIoU, which corresponds to 97.4% of the fully supervised result with the same backbone (WeCLIP-Full (Zhang et al., 2024), 81.6%). Within single-stage WSSS, it is higher than ExCEL (Yang et al., 2025b) at 77.2% with a 94.6% ratio and higher than WeCLIP (Zhang et al., 2024) at 76.4% with a 93.6% ratio. Relative to strong multi-stage methods on ResNet101, it is above CPAL (Tang et al., 2024) at 74.5% with a 95.9% ratio. The remaining gap to the fully supervised upper

bound on ViT-B* is 2.1%, indicating that our approach closes most of the gap while using weak supervision.

4.3. Ablation studies and analysis

Efficacy of Components. Within our WSSS framework, Table 5 evaluates the losses used in the proposed Feature Aligned Decoder. Supervising only the fused logits $\mathcal{L}_{\text{fuse}}$ yields 74.1%. Adding CLIP branch supervision $\mathcal{L}_{\text{clip}}$ increases the score to 77.4% (+3.3%). Supervising the diffusion branch $\mathcal{L}_{\text{diff}}$ gives 76.5% (+2.4%). Introducing the diversity loss (Yang et al., 2025b) $\mathcal{L}_{\text{div}}$ alongside $\mathcal{L}_{\text{fuse}}$ reaches 76.0% (+1.9%). Combining $\mathcal{L}_{\text{clip}}$ with $\mathcal{L}_{\text{div}}$ attains 78.5%, which is +1.1% over using $\mathcal{L}_{\text{clip}}$ alone, while combining $\mathcal{L}_{\text{diff}}$ with $\mathcal{L}_{\text{div}}$ reaches 77.4%, which is +0.9% over using $\mathcal{L}_{\text{diff}}$ alone. Using all components together gives the best performance of 79.5%, exceeding the strongest two-loss configuration $\mathcal{L}_{\text{clip}} + \mathcal{L}_{\text{div}}$ by +1.0%. These results indicate that supervising both branches improves the fused logits, and the diversity loss consistently encourages complementarity between the CLIP and diffusion branches.

Entropy-Based Weights Analysis. Fig. 8 visualizes the per-pixel weight W in Eq. (7) that combines the CLIP-based and diffusion-based CAMs. We report weights for foreground (FG) and background (BG) regions

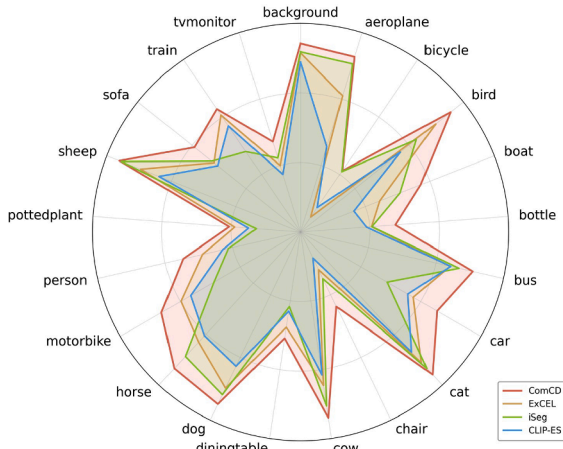


Fig. 6. Per-class mIoU on PASCAL VOC 2012. Radar plot of per-class pseudo labels mIoU on the VOC 2012 train set, comparing ComCD with CLIP-ES (Lin et al., 2023), ExCEL (Yang et al., 2025b), and iSeg (Sun et al., 2024a). A larger radius indicates higher mIoU. ComCD surpasses all three baselines on every class, demonstrating consistent gains across the 21 classes.

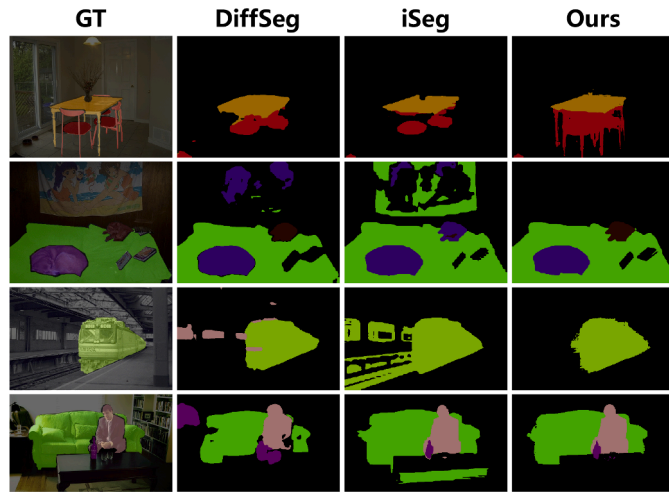


Fig. 7. Visualization of OVSS pseudo masks. We compare the results of ComCD with DiffSeg (Wang et al., 2025) and iSeg (Sun et al., 2024a).

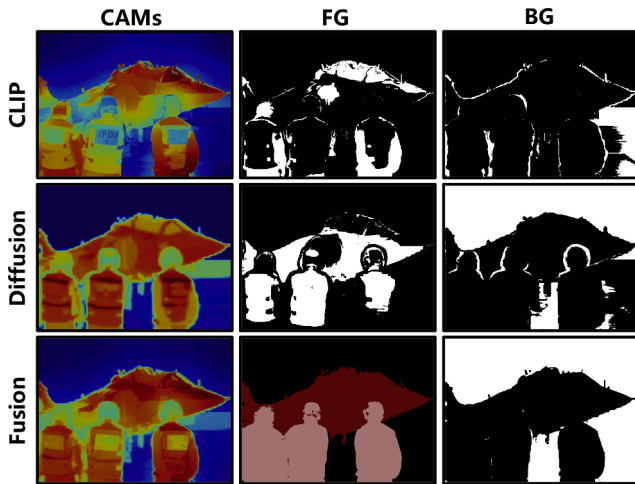


Fig. 8. Visualization of fusion weights. Per-pixel entropy-based weight W used in Eq. (7). White pixels indicate the selection of the more confident branch (CLIP-based or diffusion-based). “FG” and “BG” denote the weights computed over foreground-object regions and background regions, respectively.

Table 6

Fusion Strategy Ablation. “Seg.” denotes segmentation performance.

EBF	LGM	Seg.
✓		79.1
	✓	79.5

Table 7

Effect of fusion strategies on PASCAL VOC 2012. “Pseudo” and “Seg.” denote mIoU of pseudo masks on train and final segmentation on val, respectively.

Fusion strategy	Pseudo	Seg.
CLIP-only	74.9	76.4
Diffusion-only	75.0	75.9
Equal-weight averaging	76.3	77.1
Element-wise max	75.6	76.1
EBF (ours)	82.1	79.5

separately. White pixels indicate the selection of the more confident branch (CLIP-based CAMs or diffusion-based CAMs). Lower entropy implies higher confidence and thus a larger contribution to the fused CAM at that location. Darker pixels indicate the opposite. The visualized weights are consistent with the expected roles of the two branches. The weight maps tend to favor CLIP where class-discriminative localization is required, and diffusion where spatial coherence is beneficial. This observation aligns with prior findings that CLIP-based methods emphasize class localization (Lin et al., 2023; Xie et al., 2022; Yang et al., 2025b), whereas diffusion-based methods enhance spatial coherence (Sun et al., 2024a; Wang et al., 2025; Yoon et al., 2024a). The improvements observed in the fused segmentation masks further support the effectiveness of entropy-based weighting.

Fusion Strategy Analysis. Table 6 compares two fusion schemes at training time: the entropy-based fusion (EBF) in Section 3.2 and the Logit Gating Module (LGM). EBF first converts the CLIP-based and diffusion-based CAMs into per-pixel class distributions and then maps the entropy difference between the two branches to a scalar weight via a sigmoid, so that the branch with lower entropy (higher confidence) receives a larger contribution in the fused prediction, whereas LGM predicts a pixel-wise weight map with a 1×1 convolution applied to the concatenated logits $[S^{\text{clip}}, S^{\text{diff}}]$ followed by a sigmoid. The segmentation scores are very close (EBF 79.1%, LGM 79.5%), and we note that these results are obtained by averaging over multiple independent runs (three runs with different random seeds). This indicates that the fusion rule learned by LGM is largely consistent with the uncertainty-based weighting used in EBF rather than exploiting a completely different preference pattern between the two branches; the small but consistent gain of LGM suggests that a lightweight, learned gating can mildly calibrate the entropy-based rule in ambiguous regions, while EBF itself remains a strong and parameter-free baseline fusion mechanism.

Effect of fusion strategies. To further quantify the importance of Entropy-Based Fusion (EBF) at the pseudo-label generation stage, we compare it with four simpler fusion schemes on PASCAL VOC 2012: (1) using only CLIP-based CAMs as pseudo labels, (2) using only diffusion-based CAMs, (3) averaging the per-pixel class probabilities from the two branches (equal-weight averaging), and (4) taking the element-wise maximum of the two branches’ probabilities (max fusion). For each scheme, we first evaluate the pseudo masks on the train set and then train ComCD on these masks and report the final segmentation performance on the val set. The results are summarized in Table 7. Using only CLIP, only diffusion, or simple fusion (average / max) all stay around 75–76% pseudo-mask quality and 76–77% final segmentation, whereas EBF boosts them to 82.1% and 79.5%, respectively. This clear margin shows that our entropy-based fusion at the pseudo-label stage, which

Table 8

Impact of object size on pseudo mask quality. We partition VOC images into three subsets according to the fraction of image area occupied by foreground objects, and report mIoU on each subset.

Subset (area ratio)	ExCEL	iSeg	ComCD
Small (< 20%)	65.1	65.7	74.9
Medium (20%–50%)	77.3	77.0	84.0
Large (> 50%)	73.5	73.1	79.0

Table 9

Boundary accuracy on VOC.

We compute mIoU within a 2-pixel band around the ground-truth boundary to evaluate edge quality.

Method	Boundary mIoU
ExCEL	72.3
iSeg	66.9
ComCD	75.6

automatically selects the more trustworthy branch at each pixel, is crucial and provides substantially stronger supervision than naive combinations of the two branches.

Class-wise behaviour. Beyond the overall mIoU, we further examine the quality of pseudo masks at the class level using the radar plot in Fig. 6. Overall, ComCD tends to outperform single-branch baselines such as CLIP-ES (Lin et al., 2023), ExCEL (Yang et al., 2025b) and iSeg (Sun et al., 2024a) on most of the 21 classes, with more visible margins on several representative categories. In particular, categories with thin structures or rich local details (e.g., *chair*, *bottle*, *motorbike*, *person*) and categories that often appear with complex co-occurring backgrounds (e.g., *boat*, *sofa*) show noticeably larger radii for ComCD in Fig. 6, indicating more stable and less noisy pseudo masks. A plausible explanation is that the two branches emphasize different aspects: the CLIP branch is more sensitive to class-discriminative cues, while the diffusion branch is more effective at maintaining regional connectivity. The entropy-guided fusion appears particularly helpful in these “fine structure + strong context” scenarios, where balancing discriminativeness and spatial coherence is non-trivial for a single branch alone.

Robustness to object size and shape. To assess the behaviour of ComCD under different object scales and shapes, we partition the PASCAL VOC 2012 images (Everingham et al., 2010) into three subsets according to the fraction of image area occupied by the foreground objects, and report pseudo-mask mIoU for ExCEL (Yang et al., 2025b), iSeg (Sun et al., 2024a) and ComCD in Table 8. Across the small, medium and large subsets, ComCD achieves higher mIoU than both baselines, with relatively larger gains on small and medium objects. These subsets typically contain partially occluded instances, multiple nearby objects and irregular silhouettes, where single-branch methods often face a trade-off between preserving local details and maintaining global region consistency. The entropy-based fusion alleviates this tension to some extent by adaptively combining the two branches. For large objects that occupy a substantial portion of the image, ComCD also maintains a consistent advantage, suggesting that the proposed fusion strategy can adapt to variations in object extent and shape complexity rather than being tied to a narrow scale range.

Boundary accuracy analysis. We further introduce a boundary-focused metric to separately evaluate segmentation quality around object contours. Starting from the ground-truth masks, we construct a narrow band around each object boundary by dilating and eroding the mask by two pixels, and compute mIoU only within this band. Table 9 reports the boundary mIoU for ExCEL (Yang et al., 2025b), iSeg (Sun et al., 2024a) and ComCD. ComCD achieves the highest score among the

Table 10

Runtime, memory, and segmentation performance on VOC. We report FPS, peak GPU memory, and inference time on PASCAL VOC 2012 (val).

Method	FPS	Mem. (GB)	Time (s)	Seg.
ToCo (Ru et al., 2023) (CVPR'23)	5.93	1.87	244.22	71.1
DuPL (Wu et al., 2024b) (CVPR'24)	2.98	2.25	486.75	73.3
SeCo (Yang et al., 2024) (CVPR'24)	4.94	1.85	293.18	74.0
ExCEL (Yang et al., 2025b) (CVPR'25)	7.75	1.41	186.94	77.2
Ours	3.77	3.98	384.68	79.5

three methods, indicating fewer local gaps and less pronounced background leakage along object edges. This observation is consistent with the entropy-based weight distributions visualized in Fig. 8: in the weight maps, pixels near the boundary typically receive higher weights for the CLIP branch, whereas interior pixels tend to favour the diffusion branch. Such a division of roles between the two branches provides a more stable balance between class-discriminative boundaries and region-level coherence, which in turn helps produce smoother and more continuous contour predictions.

Failure cases and limitations. Despite the improvements observed above, ComCD still exhibits some representative failure cases. A typical scenario appears in indoor scenes containing *diningtable* and *chair* (Fig. 6): when tables and chairs are densely arranged and the background is cluttered, the diffusion branch may overly smooth the transitions between the tabletop and surrounding chairs, while the CLIP branch may focus mainly on the tabletop and underestimate thin legs and supports. For *bicycle*, the wheel spokes and frame structure are very fine; under challenging viewpoints or occlusions, both branches can have low confidence on these regions, and the fused prediction may still miss parts of the wheels or merge them with nearby background. For small or visually ambiguous *bottle* instances (e.g., transparent or highly reflective bottles), the CLIP branch sometimes emphasizes logos or high-contrast textures, whereas the diffusion branch may connect the bottle to background areas with similar colour or texture. These observations suggest that the current design, which relies on per-pixel entropy from two frozen backbones and a relatively simple fusion rule, does not explicitly model instance-level structure or richer geometric priors. We view incorporating instance-aware refinement and more expressive uncertainty modelling as promising directions for future work.

Efficiency analysis. Table 10 summarizes the runtime, memory usage, and segmentation performance on PASCAL VOC 2012. Compared with other single-stage ViT-based baselines, ComCD incurs a higher computational cost (3.77 FPS and 3.98 GB peak memory) due to the dual-branch design and feature alignment in the decoder, while achieving the best mIoU of 79.5%. In particular, our throughput remains in the same order of magnitude as prior methods (and is higher than DuPL Wu et al., 2024b), and the memory footprint is still acceptable on modern GPUs. In scenarios where segmentation accuracy is prioritized, this accuracy-efficiency trade-off is reasonable, and further reducing the overhead (e.g., via lighter diffusion backbones or feature caching) is left as promising future work.

5. Conclusion

In this work, we present ComCD, a WSSS framework that combines CLIP-based localization with diffusion-based structural coherence. First, CAMs from the two branches are fused via entropy-based weighting to produce fused CAMs, which are then converted into a pseudo mask. This pseudo mask supervises the Feature Aligned Decoder that aligns features for a shared decoder, produces logits from both branches, and applies the Logit Gating Module to yield the final fused prediction. By exploiting the complementary strengths of the two branches, the proposed fusion suppresses spurious activations, reduces background leakage, and recovers more complete objects with sharper boundaries.

CRediT authorship contribution statement

Hang Yao: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft; **Yuanchen Wu:** Formal analysis, Data curation, Writing – review & editing; **Jide Li:** Visualization, Investigation, Writing – review & editing; **Kequan Yang:** Visualization, Writing – review & editing; **Jingxin Han:** Visualization; **Xiaoqiang Li:** Resources, Supervision, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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